

Monday Warm Up: Unit 4: Inductors and capacitors, RL and RLC circuits

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1 Memory Bank

- $\tau = L/R$... The time-constant of an RL circuit.
- $\tau = RC$... The time-constant of an RC circuit.
- $v(t) = v_0 \sin(2\pi ft - \phi)$... Standard AC waveform, with amplitude v_0 , in volts, frequency f , in Hertz, time t , in seconds, and phase ϕ , in radians.
- $2\pi f_0 = 1/\sqrt{LC}$... The resonance frequency, f_0 , of an RLC circuit.
- $V_{\text{rms}} = I_{\text{rms}} Z_{\text{tot}}$... Ohm's Law for AC circuits, with total impedance Z_{tot} .

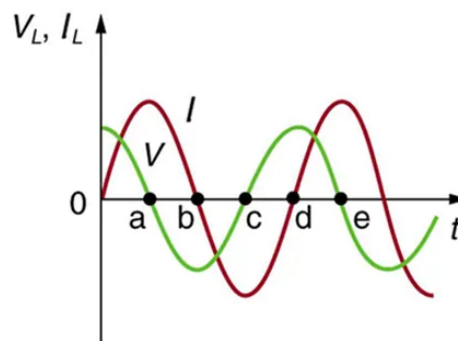


Figure 1: AC waveforms of voltage and current versus time.

2 RL Circuits

1. Your RL circuit has a characteristic time constant of 20.0 ns, and a resistance of 5.00 M Ω . (a) What is the inductance of the circuit? (b) What resistance would give you a 1.00 ns time constant, perhaps needed for quick response in an oscilloscope?

2. (a) An inductor filters high-frequency noise from power supplied to a personal computer. What inductance should it have to produce a 2.00 k Ω reactance, X_L for 15.0 kHz noise? (b) What is X_L at 60.0 Hz?

4 RLC Circuits, and Properties of AC Waveforms

2. Suppose an RC circuit is used to charge a laser for a pulse. (a) If $R = 1$ M Ω , and $C = 10$ nF, what is the characteristic time to charge the laser? (b) If we need to shorten this time, but we cannot change the capacitor, what should we do?

1. Consider the voltage waveform in Fig. 1. The voltages at points a, c, and e, are zero. Suppose that $t_a = 1$ ms, and $t_e = 3$ ms. (a) What is the period of the voltage? (b) What is the frequency of the voltage?

3 Reactance of Inductors and Capacitors

1. A 20.0 Hz, 16.0 V source produces a 2.00 mA current when connected to a capacitor. What is the capacitance?

2. What is the relative phase between the two waveforms?
3. (a) Choose an inductor and a capacitor that would give an RLC resonator a resonance frequency of 1.4 MHz. (b) If $R = 1$ k Ω , what's the total impedance at 1.4 MHz?